ToxWatch — Business Brief (Phase 1)

AI for Healthcare Graduation Project · Draft v0.2 · Synthetic data only · Workflow owners: W1 Yasmin Kamal, W2 Heba Kamal, W3 Sali Ali, W4 Reem Saleh, W5 and W6 Rania Sabri

1. Business problem

Where it sits in the patient journey: admission and triage, and the first diagnostic decisions, for acutely poisoned patients in emergency departments.

Who is affected: emergency doctors and residents who must decide quickly what to do for a poisoned patient, and the patients and families who depend on those decisions.

What it costs today:

- One Egyptian university poison centre (Ain Shams) treated **21,492 patients in 2019**; 6.2% were severe, which is about **1,330 severe cases in one centre in one year**. Aluminium phosphide poisoning carries 40–80% mortality with no specific antidote. (Sources: *ToxWatch — Egypt Market Research (2026)*, citing Ain Shams PCC 2019 and the Egyptian J. Forensic Sciences 2025.)
- A second centre agrees on scale: Tanta University's toxicology centre received **9,713 cases between 2017 and 2021**, rodenticides being the most frequent cause (same source page, citing PMC9896829).
- Those figures come from two centres and are 5 to 7 years old. They show scale, not the national burden, which is under-recorded.
- **Hypothesis, not yet confirmed:** a doctor in a governorate hospital at night often has no toxicologist to ask, and the current workaround is a phone call or a general-purpose chatbot that gives uncited answers. **Not validated in this project:** no interviews were carried out. Confirming this with emergency doctors and pharmacists is the first step of a real pilot (see the pitch's next steps).

2. Problem statement (Template A1)

Emergency doctors and residents in Egyptian governorate hospitals struggle to get a fast, referenced first-line toxicology answer when a poisoned patient arrives and no toxicologist is reachable. Today this causes delay and uncertain decisions in cases where early management changes outcome (impact to be quantified in the Phase 4 timing test). The current workaround is calling a university poison centre, if someone answers, or asking a general chatbot, which falls short because it depends on reaching a person, leaves no structured record, and gives answers without sources or protection for patient identifiers. Solving it would give doctors a cited, safety-checked draft answer in minutes, route dangerous requests to a human, and give the hospital an auditable record.

3. Business objectives

Targets below are proposed by the team. Baselines are measured in Phase 4, not assumed.

1. Reduce the time from a clinician's question to a cited first-line answer from **about 15 minutes to under 2 minutes**. The 15 minutes is the team's estimate from practice, to be checked by timing 10 synthetic cases by hand.
2. **100%** of critical-tier, self-harm and lethal-dose requests are refused or held for human approval; none is answered automatically (tested on synthetic cases).
3. **At least 90%** of answers carry a source citation or say "insufficient information"; none states an unsupported dose.

4. Target audience

Group	Who	Notes
User (operates the system)	Emergency physicians and residents; clinical pharmacists	Main user
Beneficiary	Poisoned patients and their families	Never see the system directly
Buyer (decides and pays)	Hospital medical director or head of the emergency department	Hypothesis; to be confirmed in pricing conversations

Persona — Dr. Mona, emergency resident, governorate general hospital (synthetic persona). Works night shifts with one senior on call. Today she looks up doses in a book or phones a poison centre while the patient waits. She is frustrated by answers with no source and by typing patient names into tools she does not trust.

5. Success metrics (Template A4)

KPI	Lens	Baseline today	Target	How it is measured	Actual result
Minutes from question to cited first-line answer	Cost	About 15 min (team estimate from practice, not measured)	Under 2 min	Stopwatch on the same 10 cases with and without the system	Not measured
Dangerous requests refused or held for a human	Outcomes	No enforced rule today; general chatbots answer without a human check	100%	10 synthetic unsafe requests (self-harm, lethal dose, injection)	3 of 3 held; 1 of 10 cases run
Answers with a citation or "insufficient information"	Outcomes	Not guaranteed today; general chatbots give uncited answers	At least 90%	LLM-as-judge groundedness score, checked by two humans	Not established (humans 3.75 of 5)
Clinicians who say the draft saves them effort	Workforce well-being	Not applicable before the system; asked in the survey after testing	4 of 5	Short survey after testing	Survey not run

6. Scope

In scope: a clinician types a toxicology question (adult or paediatric, English or Arabic); identifiers are masked; the system classifies risk, retrieves approved reference documents, calculates dose arithmetic with a tool, drafts a cited answer, and routes critical cases to a human. Every run is logged and traced; threshold breaches raise an alert.

Out of scope: diagnosis and treatment decisions, hospital record-system links, real patient data, poison-centre integration, and the Telegram voice-to-SOAP flow (in the prototype, not in this demo).

The AI's role is assistive. The AI drafts; a clinician decides.

The system must never: make a final medical decision, help anyone harm themselves or others, invent a dose or citation, release a critical-tier answer without a human's approval, or pass real patient identifiers to a model.